

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS student (NC State, May 2026) with 0-2 years production experience shipping full-stack features end-to-end in TypeScript, React, and Node.js. At Isomer AI, owned real-time event pipelines (Pusher), document processing systems, RBAC, and multi-model LLM integrations across 7 production deployments -- 394 commits, zero rollbacks. Built AlgoPulse, re-architecting a distributed AI backend to handle 2,000 async requests/second using Temporal.io durable workflows, Kafka message queuing, and WebSocket streaming -- the exact distributed systems patterns Lindy needs at scale. Strong bias toward shipping: 3 hackathon MVPs built in 48-hour sprints, consistently top placements.

CORE COMPETENCIES

TypeScript & React

Node.js / Next.js

Real-Time Systems (WebSockets / Pusher)

Distributed Systems (Kafka / Temporal)

LLM API Integration (MCP / Multi-Model)

PostgreSQL & Redis (Caching)

Third-Party API Integration

End-to-End Feature Ownership

Python & AI Pipelines

EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern - Remote

- Owned **real-time event pipeline** end-to-end: designed and built Pusher-based server-side event emission in Node.js + client-side React subscription layer with RBAC controls -- shipped across 7 production deployments with zero rollbacks
- Built backend pipelines for document processing (PDF generation, email parsing via Microsoft Graph API, S3 storage), integrated Auth0 for auth and Redis for session caching across the full TypeScript/Node.js/PostgreSQL stack
- Integrated **multi-model LLM tooling** via MCP across Claude, GPT-4o, and Gemini; wired Temporal.io for durable AI workflow orchestration -- **394 commits**, high code review participation

NC State University Apr 2025 - Present

Graduate Research Assistant

- Built a **React + JBrowse 2** genomic data platform on Google Cloud Run serving 100+ researchers across 500GB of datasets; applied ML hyperparameter tuning improving model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ Python automation workflows integrating third-party APIs and ML models; deployed on AWS & GCP with Grafana monitoring; improved health claim accuracy by 25%

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Built **Kafka streaming + Spark batch pipelines** for real-time support ticket analysis; reduced latency by 20% and processing time by 15%

PROJECTS

AlgoPulse - Distributed AI Platform Independent · 2025

Identified a polling bottleneck under load and re-architected a full-stack AI platform to event-driven streaming: **Temporal.io** for durable workflow state with automatic retries on failure, **Kafka** for worker coordination and guaranteed message delivery, Supabase Edge Functions for real-time WebSocket streaming to clients. Scaled to **2,000 async AI inference requests/second** -- zero dropped messages under fault injection.
TypeScript · React · Node.js · Temporal.io · Kafka · WebSockets · Supabase · PostgreSQL

CryptoKnight - Trustless AI Trading Agent HackNC State · 2025

Built a **Next.js 16 + Node.js** full-stack app integrating ERC-4337 Account Abstraction APIs, Alchemy Account Kit, and a Python agent backend. Owned the complete feature end-to-end -- from smart account session key policy UI to autonomous on-chain trade execution -- in a 48-hour sprint.

Next.js 16 · Node.js · TypeScript · viem · wagmi · Python · ERC-4337

Glance AI - Privacy-First RAG System Independent · 2025

Built a Python RAG pipeline with per-tenant data isolation at the vector retrieval layer for fintech/healthcare compliance environments -- 0 cross-tenant context leakage by design.

Python · LLMs · Vector Databases · RAG

PerCent - GenAI Finance App HackNC24 Top 10 of 100+ teams

Shipped a complete full-stack app -- goal tracking, expense logging, RAG-powered AI chatbot -- in 48 hours. Top 10 of 100+ teams at HackNC24.

React · Node.js · Python · Hugging Face · RAG · Selenium

EDUCATION

North Carolina State University - MS Computer Science	Aug 2024 - May 2026
Courses: Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms	
Mumbai University - BTech Computer Science	Aug 2020 - May 2024
CGPA: 9.32/10 · Courses: DS & Algorithms, Machine Learning, Software Engineering, Big Data Infrastructure	

SKILLS

- Frontend / Full Stack:** TypeScript, React (v16+), Next.js, Node.js, JavaScript (ES6+), GraphQL, Tailwind CSS, HTML/CSS
- Backend & APIs:** Express, PostgreSQL, Redis, Prisma, REST APIs, WebSockets, Pusher, Auth0, S3
- Distributed Systems:** Kafka, Temporal.io, Spark, Docker, CI/CD, AWS, GCP, Git
- AI/LLM:** OpenAI API, Anthropic Claude, MCP, LangChain, RAG, PyTorch, TensorFlow, Hugging Face
- Other:** Python, Java, C, SQL, Grafana, Sentry